Picard-Fuchs Equations and Cusps Forms [DRAFT]

Ryan Lam

May 10, 2023

Abstract

This paper is written for the purpose of MATH 492 Directed Research Unit at the University of Arizona. Supervised by Dr Douglas C Haessig.

Contents

1 Introduction **2**

2 Preliminary Knowledge **2**
2.1 The Picard-Fuchs Equation 3
2.2 Monodromy Representation of an ODE 6

3 Example: The Legendre Family **7**
3.1 Monodromy of solutions to the equation (2) 8
3.2 Monodromy of the Lambda Modular Function 8

4 Producing Cusps Forms **11**

5 Generalization and Future Directions **12**
5.1 The $p = 3$ case 12
5.2 Conclusion and Further Work 14

Bibliography **15**

List of SageMath Codes

1 Defining the function `find_Picard`. 5
2 Computing the Wronskian of (2). 7
3 The first 25 terms of the q-expansion of λ stored as a list in the variable `LamFunc_Coeff`. 11
4 Establishing Linear Relationship between functions $G_i^{(k)}$ and basis of $\mathbb{S}_{k+2}(\Gamma_1(4))$. . 13
5 The variable `x_2` which holds the first 99 terms of the q-expansion of x_2. 14

THIS IS A DRAFT COPY. If I have time, I will tikz the pictures. The red parts indicate what I will potentially write up in the near future - probably will not.

1 Introduction

The intricate connection between elliptic curves (defined over $\mathbb{Q}$) and modular forms has long been studied, with Modularity Theorem being the most celebrated result, which allows Wiles to prove the Fermat last theorem in Wiles [1995]. In this article, we look at one such connection. With a choice of a suitable family of elliptic curves E_t, an associated modular form m and a certain modular group G, we show, with explicit coefficients, that we can use an expression to determine a basis for the cusps forms of weight k of G, denoted as $\mathbb{S}_k(G)$. Recall that

Definition 1.1. A **modular form of weight** k **on** Γ, where Γ is a subgroup of $\mathrm{PSL}_2(Z)$, is a holomorphic function f from the upper half plane $\mathbb{H}$ to the complex numbers $\mathbb{C}$, with the requirement that for all $\tau \in \mathbb{H}$:

$$f\left(\frac{a\tau+b}{c\tau+d}\right) = (c\tau+d)^k f(\tau) \qquad \text{for all } \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma.$$

If f is a modular form of weight 0 on Γ, we will simply refer to it as a **modular form on** Γ.

This article provides new computational details, with `SageMath` code attached at relevant sections.

The choice of E_t, m, and G is far from random. Finding the right triples require computing the Picard-Fuchs equation of E_t and some knowledge of monodromy representations, which is reviewed in section 2. One classical example is presented in Yang [2019], where E_t is the Legendre family of elliptic curve $L_t := \{(x, y) : y^2 = x(x-1)(x-t)\}$, m is the lambda modular function λ, and $G = \Gamma_1(4)$. We will present this example in a more friendly language in section 3 and 4. The compatibility requirement is the monodromy of the Picard-Fuchs Equation of E_t and m, which we will mention in subsection 3.2. Lastly, we present the difficulty to generalize the argument to the family of elliptic curves presented in [Haessig, 2021, Theorem 2.1] in section 5.

2 Preliminary Knowledge

Our main interests are elliptic curves, for technical purposes, we will define it over a characteristic 0 field K, usually taken to be $\mathbb{C}$.

Definition 2.1. An **elliptic curve** E (over K) is a (zero) set $E := \{(x, y, z) : F(x, y, z) = 0\}$ of a degree 3 homogeneous complex polynomial $F \in \mathbb{C}[x, y, z]$.

Note that F can be "de-homogenize" to a function in two variables,

$$f(x, y) := F(x, y, 1)$$

Theorem 2.2. Any elliptic curve E has form of $y^2 = x^3 + ax + b$ for some $a, b \in \mathbb{C}$ with $4a^3 + 27b^2 \neq 0$. This is called the **normal form** of E.

Proof. The proof is found in any ordinary text on elliptic curves such as [Silverman, 2014, Sect. III.1]. □

One important consequence is that in any elliptic curve realized as an algebraic curve, the polynomial $x^3 + ax + b$ is always irreducible and have three distinct roots.

Moreover, if the elliptic curve is defined over $\mathbb{C}$, it can be put on a Riemann surface (one-dimensional complex manifold) structure. A complete treatment of this topic can be found in [Donaldson, 2011, Chapter 3]. We present one consequence of realizing E as the zero set of $\mathbb{CP}^2$:

Theorem 2.3. Any elliptic curve can be seen as a quotient topological space $\mathbb{C}/L$, where L is a (non-degenerate) lattice $L := \text{span}_{\mathbb{Z}}\langle\omega_1, \omega_2\rangle, \omega_2/\omega_1 \notin \mathbb{R}$. The elliptic curve (Riemann surface) isomorphism is given by $\mathbb{C}/L \mapsto \mathbb{CP}^2$:

$$z \mapsto [1 : \wp'(z) : \wp(z)], \qquad 0 \mapsto [0 : 1 : 0]$$

where $\wp$ is the **Weierstrass $\wp$-function**:

$$\wp(z) := \frac{1}{z^2} + \sum_{\lambda \in L \setminus \{0\}} \left(\frac{1}{(z-\lambda)^2} - \frac{1}{\lambda^2} \right).$$

Sometimes, we also brought it down to $y^2 = f(x)$, where f is a degree 3 polynomial in x, this is sometimes referred to as the "long" normal form.

2.1 The Picard-Fuchs Equation

We want to consider a family of elliptic curves E_t controlled by a parameter $t \in \mathbb{C}$. That is, consider a level set of a family of homogeneous degree 3 polynomials $F_t(x, y, z)$. We can "de-homogenize" F_t and write E_t into normal form, namely, let:

$$y^2 = x^3 + a(t)x + b(t)$$

be the elliptic curve E_t, where $a(t), b(t) \in \mathbb{C}(t)$.
For each t, There is a holomorphic 1-form ω we are interested in.

Definition 2.4. Fixing t, we define the following form for each elliptic curve E_t:

$$\omega(t) = \frac{dx}{y} = \frac{dx}{\sqrt{x^3 + a(t)x + b(t)}},$$

for which we will write $\omega = g(x, t)\, dx$.

Here is a theorem motivating this choice of $\omega(t)$:

Theorem 2.5. The inverse of the isomorphism in theorem 2.3 is as follows, the lattice L can be found by the integral

$$\omega_1 = \int_{p_1} \omega, \qquad \omega_2 = \int_{p_2} \omega,$$

where p_1, p_2 are the basis of the torus representing E, illustrated in figure 1.

Note that we can differentiate ω with respect to t as

$$\frac{d\omega}{dt} := \frac{\partial g(x,t)}{\partial t} dx.$$

And hence $\dfrac{d\omega}{dt}$ is still a 1-form.

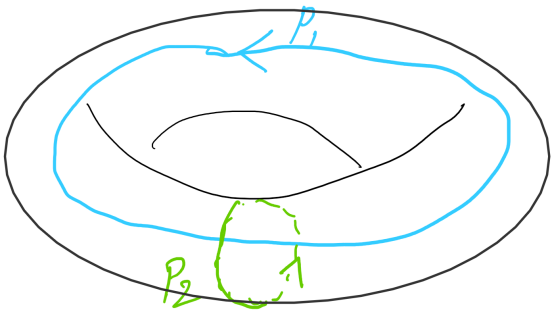


Figure 1: The curves p_1 and p_2.

Definition 2.6. The **Picard-Fuchs equation associated to** E_t, is the second-order homogeneous ordinary differential equation (ODE) where $\pi(t) = \int_\gamma \omega(t)$, where γ is any 1-chain. It has the form o

$$\frac{d^2\pi}{dt^2} + B\frac{d\pi}{dt} + C\pi = 0 \tag{1}$$

where B, C are functions of t.

For different families of elliptic curves E_t, the Picard-Fuchs equation will be different, we are interested in its solution's monodromy, which we will define in the next section.

The method of calculating the Picard-Fuchs Equation is outlined in the document by Schnell Schnell [2004], we have produced the relevant `SageMath` function `find_Picard` as given in Listing 1.

The author also posted a short article explicitly computing the intermediate steps at Lam [2023]. Note that, in all of our code, we initiate the variables using the following line.

```
x, t, A_0, A_1, A_2, A_3, A_4, B, C, q =var('x t A_0 A_1 A_2 A_3 A_4 B C q')
```

In particular, the quantity `x, t, q` will be always be considered as variables, and `B, C` are functions of `t` if they are solved in the `find_Picard` function.

Given a period $\omega(t) = \dfrac{dx}{y(t)}$, we can input $y(t)$ as the argument `y` and the function `find_Picard(y)` will print out the required expression B, C as a function of t. For example, we can recover the example presented in Schnell's document:

$$E_t := t(x^3 + y^3 + z^3) - 3xyz$$

written in (long) normal form, we have

$$E_t : y^2 = tx^3 + \frac{1}{4}\left(1 - 4t^3\right)x^2 + \frac{1}{3}t^2\left(t^3 - 1\right)x - \frac{1}{27}t\left(t^3 - 1\right)^2$$

Hence the 1-form ω is:

$$\omega(t) = \frac{dx}{y(t)} = \frac{1}{\sqrt{tx^3 + \frac{1}{4}\left(1 - 4t^3\right)x^2 + \frac{1}{3}t^2\left(t^3 - 1\right)x - \frac{1}{27}t\left(t^3 - 1\right)^2}}\,dx$$

```
def find_Picard(y, index_fix=0):
    """
    Input elliptic curve y in y = sqrt(...), where
    ... is in normal form, display the coefficient B, C for
    the Picard-Fuchs Equation.
    """
    f = 1/y
    f_dot = f.diff(t)
    f_ddot = f_dot.diff(t)
    eta = f_ddot+B*f_dot+C*f
    eta = eta.factor()
    f_0 = A_0*(f^3)
    f_1 = A_1*x/(y^3)
    f_2 = A_2*x^2/(y^3)
    f_3 = A_3*x^3/(y^3)
    f_4 = A_4*x^4/(y^3)
    d_0 = f_0.diff(x)
    d_1 = f_1.diff(x)
    d_2 = f_2.diff(x)
    d_3 = f_3.diff(x)
    d_4 = f_4.diff(x)
    find_As_exp = eta+d_0+d_1+d_2+d_3+d_4
    find_As_exp = find_As_exp.factor()
    # index fix
    find_As_poly = find_As_exp.operands()[index_fix]
    coe_find_A = find_As_poly.coefficients(x, sparse=False)
    # Coefficient of x^2, ... x^6 = 0
    A_Coeff = solve(coe_find_A[2:7], A_0, A_1, A_2, A_3, A_4)
    x1_coeff = coe_find_A[1]
    x0_coeff = coe_find_A[0]
    one_solve = x1_coeff.subs(A_Coeff)
    zero_solve = x0_coeff.subs(A_Coeff)
    sol = solve([one_solve, zero_solve], B, C)
    func_B = sol[0][0]
    func_C = sol[0][1]
    return [func_B, func_C]
```

Listing 1: Defining the function `find_Picard`.

then running

```
Ex1 = (sqrt(t*(x^3)+(1/4)*(1-4*t^3)*(x^2)+(1/3)*
           (t^2)*(t^3-1)*x-(1/27)*t*(t^3-1)^2))
find_Picard(Ex1, index_fix=1)
```

Would yield the result

$$\left[B = \frac{4\,t^3 - 1}{t^4 - t}, C = \frac{2\,t}{t^3 - 1}\right].$$

as presented in the document of Schnell. Note that we set the parameter `index_fix` to 1. This is a technicality to ensure `SageMath` always pick the polynomial we need for the `.coefficeints` attribute to function correctly. There is currently no known way to decide which number of `index_fix` would produce the result. But to the author's knowledge and working with examples, it is most likely to be 0 (by default) or 1.

2.2 Monodromy Representation of an ODE

More generally, it turns out that for any elliptic curve family E_t, equation (1) is an ODE with regular singularities at $0, 1, \infty$. From here, we can use tools from Fuchsian equations to investigate it. We adapt the notations introduced in Kuga [1993]. Consider the solutions of the ODE (1) on $D := \mathbb{C} - \{0, 1, \infty\}$, it is a two-dimensional vector space over the complex number, which we will denote as $V_\#$. However, one can consider the universal covering surface and the solution of (1) on there. We now recall some algebraic topology results about covering surfaces:

Definition 2.7. Given $f : D' \to D$ a covering. A **deck/covering transformation** is a homeomorphism $\sigma : D' \to D'$ such that the following diagram commutes:

$$\begin{array}{ccc} D' & \xrightarrow{\sigma} & D' \\ & {\scriptstyle f}\searrow \quad \swarrow{\scriptstyle f} & \\ & D & \end{array}$$

If $f : \widetilde{D} \to D$ is the universal covering, the set of all covering transformations is denoted by $\mathrm{Aut}(\widetilde{D}/D)$, and it is a group under composition.

Theorem 2.8. [Hatcher, 2002, Proposition 1.39] $\mathrm{Aut}(\widetilde{D}/D) \cong \pi_1(D)$, the (first) fundamental group of D.

In particular, the notion of monodromy representation is important to us,

Theorem 2.9. [Kuga, 1993, Theorem 16.3] If $w \in V_\#, \gamma \in \mathrm{Aut}(\widetilde{D}/D)$, then the pullback of w by γ, that is, $\gamma^* w$, is also a solution of the ODE (1). That is, $w \in V_\#$.

Hence, by choosing a basis $\mathrm{span}_{\mathbb{C}}\langle w_1, w_2\rangle$ of $V_\#$, any element in the fundamental group (That is, $\mathrm{Aut}(\widetilde{D}/D)$) give rises to a linear map in $V_\#$, sending an element by its pullback by γ.

Definition 2.10. The above map is a homomorphism $\rho : \mathrm{Aut}(\widetilde{D}/D) \to \mathrm{Hom}(V_\#, V_\#)$, That is, it is a representation of $\mathrm{Aut}(\widetilde{D}/D)$ on $V_\#$. ρ is called the **monodromy representation** of equation (1).

Consider $D = \mathbb{C} - \{0, 1, \infty\}$, then by [Iwasaki et al., 2012, p. 79], the generators of Γ can be chosen to be two curves which we name γ_0, γ_1. They are two positively oriented curves, winding once around 0 and 1 respectively, based on any point chosen in D. An illustration is shown in figure 2.

Hence, our matrix of interest would be $\rho(\gamma_0)$ and $\rho(\gamma_1)$, and we would also compute the image of ρ.

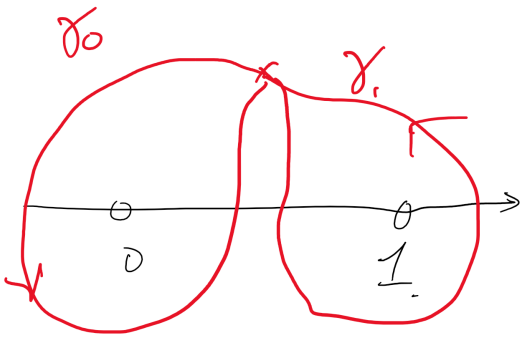


Figure 2: A choice of the curve γ_0 and γ_1.

3 Example: The Legendre Family

In this section, our main example is the **Legendre family of elliptic curves**:

$$E_t : \qquad y^2 = x(x-1)(x-t)$$

Putting `sqrt(x*(x-1)*(x-t))` through the `find_Picard` function with `index_fix=0` yields the ODE:

$$\frac{d^2\pi}{dt^2} + \frac{2\,t-1}{t^2-t}\frac{d\pi}{dt} + \frac{1}{4\,(t^2-t)}\pi = 0 \tag{2}$$

This is a hypergeometric ODE. Moreover we can compute its Wronskian:

```
B_Len = (2*t-1)/(t*t-t)
C_Len = 1/(4*(t*t-t))
W_Len = (B_Len)*diff(C_Len, t)-(C_Len)*diff(B_Len, t)
```

Listing 2: Computing the Wronskian of (2).

And printing out `W_Len` we see that

$$W = -\frac{1}{2\,(t-1)^2 t^2} \tag{3}$$

Moreover, the solution w of (2), according to the theory of Fuchsian differential equation, is a linear combination of two linearly independent solutions w_1, w_2 where:

- w_1 is holomorphic at $z = 0$, and
- w_2 is holomorphic at $z = 1$

and the solution has a logarithmic singularity. More details can be found at [Kuga, 1993, Theorem 18.6].

3.1 Monodromy of solutions to the equation (2)

Following [Bateman, 1953, p.95], we choose a basis $\pi_1(t) = {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; t\right), \pi_2(t) = i \cdot {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; 1-t\right)$.

Theorem 3.1. With a branch choice of log of $|\arg(-z)| < \pi$, equivalently $0 < \arg(z) \leqslant 2\pi$, yet equivalently the complex plane minus the positive real axis, we have:

$$\rho(\gamma_0) = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}, \qquad \rho(\gamma_1) = \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix}$$

Equivalently, under γ_0, π_1 is analytically continued to π_1, and π_2 is analytically continued to $2\pi_1 + \pi_2$.

Proof. This is a result in [Yoshida, 2013, Page 98] with $c_1, c_2, c_3 = (-1, -1, -1)$. □

However, there is another proof by directly manipulating the identity

$$\frac{\pi}{2}\pi_1(z) + \frac{1}{2i}\log(1-z)\pi_2(z) = \sum_{n=0}^{\infty} \frac{\left(\frac{1}{2}\right)_n \left(\frac{1}{2}\right)_n}{n!n!} \left[\psi(n+1) - \psi\left(n + \frac{1}{2}\right)\right](1-z)^n$$

where $\psi(z) := \dfrac{\Gamma'(z)}{\Gamma(z)}$ is the logarithmic derivative of the Gamma function. Analytic continuation details, explicit series computation

3.2 Monodromy of the Lambda Modular Function

Now we motivate a choice of a modular form m. Note that we can view any elliptic curve in $\mathbb{CP}^2$, the complex project plane as an algebraic curve. Hence, any elliptic curve is a Riemann surface.

Theorem 3.2. Any elliptic curve is isomorphic (as Riemann Surface) to one in the form

$$y^2 = x(x-1)(x-\lambda)$$

for some $\lambda \in \mathbb{C}$.

Proof. The complete proof can be found at [Silverman, 2014, Proposition III.1.7], but there is an important idea: we have seen that any elliptic curve $y^2 = f(x)$, f will have three distinct roots $y^2 = (x-e_1)(x-e_2)(x-e_3)$, then there is an isomorphism by changing coordinate to get $\lambda = \dfrac{e_3 - e_1}{e_2 - e_1}$. □

The above idea is also relevant to an elliptic curve E defined over $\mathbb{C}$, where

$$e_1 := \wp\left(\frac{\omega_1}{2}\right), e_2 := \wp\left(\frac{\omega_2}{2}\right), \text{ and } e_3 := \wp\left(\frac{\omega_1 + \omega_2}{2}\right). \tag{4}$$

where $E = \mathbb{C}/L, L := \text{span}_{\mathbb{Z}}\langle\omega_1, \omega_2\rangle$.

It turns out that setting $\tau = \dfrac{\omega_2}{\omega_1}$, due to the properties of Weierstrass $\wp$-function. Hence we define

Definition 3.3. The **modular lambda function** λ is a function from $\mathbb{H} \to \mathbb{C}$:

$$\lambda(\tau) = \frac{e_3 - e_1}{e_2 - e_1}$$

where e_1, e_2, e_3 are as (4), but scaled according to $\tau = \frac{\omega_2}{\omega_1}$. That is:

$$e_1 := \wp\left(\frac{1}{2}\right), e_2 := \wp\left(\frac{\tau}{2}\right), \text{ and } e_3 := \wp\left(\frac{1+\tau}{2}\right). \tag{5}$$

It turns out that this is a modular form we need, due to the following important properties of λ:

Proposition 3.4. [Ahlfors, 1979, Ch. 7, Theorem 7]

- λ is a modular form on the following group

$$\Gamma(2) := \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \mod 2, \text{elementwise} \right\}.$$

- λ extends to a universal covering map of $\overline{\mathbb{C}} - \{0, 1, \infty\}$, by mapping $0 \mapsto 1, 1 \mapsto \infty, \infty \mapsto 0$ and choosing the domain, often referred as the **fundamental domain of** λ, to be the following subset in $\mathbb{H}$:

$$\mathcal{F}_\lambda = \left\{ \tau : \left(\mathrm{Re}(\tau) \in (-1, 1) \text{ and } \min\left(\left| \tau - \frac{1}{2} \right|, \left| z + \frac{1}{2} \right| \right) > \frac{1}{2} \right) \text{ or } \mathrm{Re}(\tau) = -1 \text{ or } \left| \tau + \frac{1}{2} \right| = \frac{1}{2} \right\}.$$

The fundamental domain is illustrated in figure 3, note that we divided it into equal halves with the boundary A, B, C labelled. Note that as λ is real at the imaginary axis, an application of Schwartz's reflection principle would make the boundary B reverse its orientation.

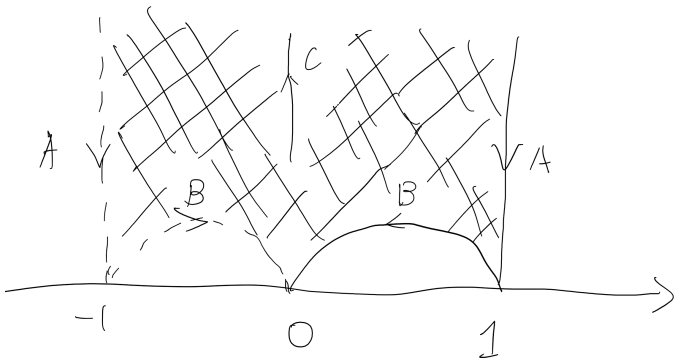


Figure 3: The set $\mathcal{F}_\lambda$.

λ can also be analytically continued along the curves γ_0, γ_1. Note that λ will map the boundaries of $\mathcal{F}_\lambda$ into the real axis as shown in figure 4.

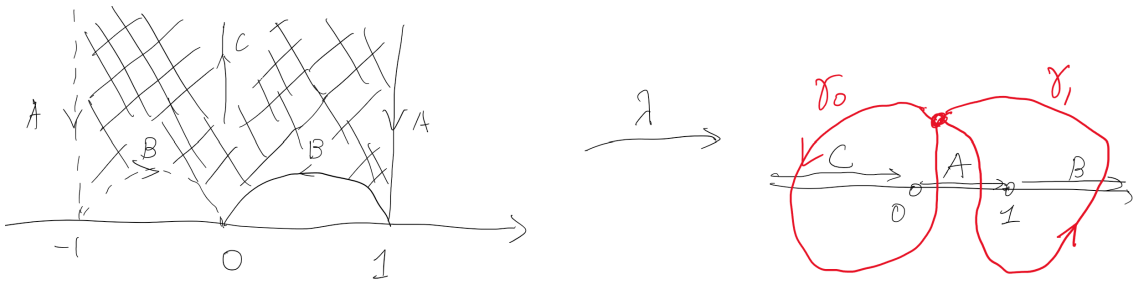


Figure 4: The image of the boundaries of $\mathcal{F}_\lambda$ under λ.

Theorem 3.5. Analytically continuing λ under the curves γ_0, γ_1 will produce:

$$\lambda \mapsto \lambda + 2, \qquad \lambda \mapsto \frac{\lambda}{1-2\lambda}$$

respectively.

Sketch. This can be done graphically by considering how a path can be pull back to $\mathbb{H}$, as the universal covering space of $\overline{\mathbb{C}} - \{0, 1, \infty\}$. Ensuring the boundaries orientation are correct throughout the process gives the required transformation. A graphical illustration is at figure 5 below. □

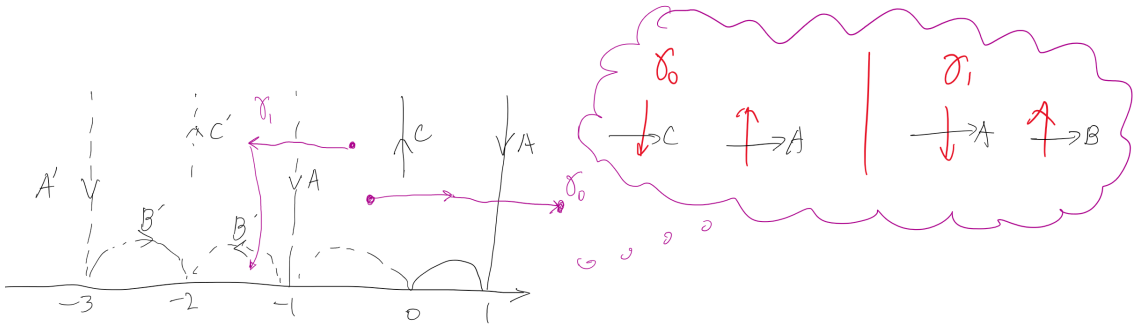


Figure 5: A Figure illustrating the main ideas of proof of Theorem 3.5.

Note that on $\mathbb{H}$ there is a natural action by $\mathrm{SL}_2(\mathbb{C})$ with $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ identified as the map $z \mapsto \frac{az+b}{cz+d}$. That means the two matrices in Theorem 3.1 and 3.5 are the same. Generally, with a case which we will briefly discuss in section 5, the requirement for a choice of modular form m and a family of elliptic curves E_t, is that the monodromy of the Picard-Fuchs equation of E_t and analytically continuation of m need to match up.

4 Producing Cusps Forms

In order to produce the linear relationship mentioned in section 1, one explicit way is to compare the q-expansion of a modular form.

Theorem 4.1. Let k be an even natural number, then every modular form of weight k on any group Γ f admits what is called a **q-expansion**, that is, a series expansion

$$f(\tau) = \sum_{n=0}^{\infty} a_n e^{2\pi i \tau n}$$

Proof. This is similar to the proof of the existence of Fourier series. □

Definition 4.2. A **cusp form of weight** k on G is a modular form of weight k on G, with the constant term of q-expansion being 0.

The q-expansion of λ can be easily found online (for example, at Somos [2015]), we record their first few terms for computational purposes.

```
LamFunc_Coeff = [0, 16, -128, 704, -3072, 11488, -38400, 117632, -335872, 904784,
-2320128, 5702208, -13504512, 30952544, -68901888, 149403264, -316342272,
655445792, -1331327616, 2655115712, -5206288384, 10049485312, -19115905536,
35867019904, -66437873664]
```

Listing 3: The first 25 terms of the q-expansion of λ stored as a list in the variable `LamFunc_Coeff`.

Continuing with the Legendre family example in section 3 with Picard-Fuchs Equation (2). Note that both the group $\Gamma(2)$, where λ is a modular form; and the group (proof)$\Gamma := \left\langle \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}, \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix} \right\rangle$, the generator of the deck transformation group, are conjugate to the group

$$\Gamma_1(4) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}) : a, d \equiv 1 \pmod 4, c \equiv 0 \pmod 4 \right\}.$$

So a good candidate for the cusps forms maybe $G = \Gamma_1(4)$. Now, we want to compare the q-expansion of cusps forms $\mathbb{S}_M(\Gamma_1(4))$ and the solution of (2) for any even integer M. Firstly, we need a fact of the cusps forms on G (provided without proof).

Theorem 4.3. For any even natural number M, $\mathbb{S}_M(\Gamma_1(4))$ can be realized as a set of q-expansions, which is a $\mathbb{Z}$-module (hence, it can be seen as a sequence space with integer coefficients). The dimension of $\mathbb{S}_{k+2}(\Gamma_1(4))$, is $\frac{k}{2} - 1$ for even k, 0 otherwise.

Now we establish an explicit linear relationship between the solutions of the Picard-Fuchs equation and the basis. To start we consider the following expression: Define for $k \geqslant 4$ even, $1 \leqslant i \leqslant \frac{k}{2} - 1$:

$$G_i^{(k)}(q) = (F(\lambda(q)))^{k+2}(1 - \lambda(q))(\lambda(q))^i \tag{6}$$

where

$$F(t) = {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; t\right). \tag{7}$$

Theorem 4.4. The set $\left\{G_i^{(k)} : 1 \leqslant i \leqslant \frac{k}{2} - 1\right\}$ forms a basis for $\mathbb{S}_{k+2}(\Gamma_1(4))$

This is well known, and we establish the explicit coefficients by computing both of their q-expansion on `SageMath` in Listing 4. We obviously need the expression `B_Len, C_Len, LamFunc_Coeff` at Listing 2 and 3. For some reason unknown to the author, the first line in Listing 4 is required in order for the Power Series ring to achieve a correct coercion behavior.

The above functions `G` and `Gamma14_get_qexp` allow us to print out the q-expansions up to specific powers of q specified by the parameter `n` (default at 15). For example, running

```
Gamma14_get_qexp(6);
G(6);
```

Prints out the following:

```
Gamma1(4) Basis #1 of S_8: up to q^15
q + 12*q^3 - 210*q^5 + 1016*q^7 - 2043*q^9 + 1092*q^11 + 1382*q^13 + O(q^15)


Gamma1(4) Basis #2 of S_8: up to q^15
q^2 - 8*q^4 + 12*q^6 + 64*q^8 - 210*q^10 - 96*q^12 + 1016*q^14 + O(q^15)


Power expansion up to q^15 for G_1, k=6

16*q + 128*q^2 + 192*q^3 - 1024*q^4 - 3360*q^5 + 1536*q^6 + 16256*q^7 + 8192*q^8 - 32688*q^9 -
26880*q^10 + 17472*q^11 - 12288*q^12 + 22112*q^13 + 130048*q^14 + O(q^15)


Power expansion up to q^15 for G_2, k=6

256*q^2 - 2048*q^4 + 3072*q^6 + 16384*q^8 - 53760*q^10 - 24576*q^12 + 260096*q^14 + O(q^15)
```

Then, comparing coefficients, if we write c_1, c_2 for a basis of $\mathbb{S}_8(\Gamma_1(4))$, we have

$$\begin{cases} G_1^{(6)} = 16c_1 & +128c_2 \\ G_2^{(6)} = & 256c_2 \end{cases} \tag{8}$$

The process is similar for other even integers k.

5 Generalization and Future Directions

5.1 The $p = 3$ case

Our aim is to produce a similar equation to (8) using the elliptic curve family

$$E_t : \qquad y^2 = x^3 + x^2 - t \tag{9}$$

```
x, t, B, C, q =var('x t B C q')
F_list = (hypergeometric([1/2, 1/2], [1], t).series(t,
        50)).coefficients(sparse=False) # 50 terms of F extracted
t_Ring.<t> = PowerSeriesRing(QQ)
Q_Ring.<q> = PowerSeriesRing(QQ)
F_Series = t_Ring(F_list, 49)
Lam_Series = Q_Ring(LamFunc_Coeff, 25)
F = F_Series.subs({t: Lam_Series})
def G(k, n=15):
    """
    Given k an even integer,
    Prints the q-expansion of all the function G_i^(k),
    1 <= i <= k/2-1, for powers of q up to n
    """
    L = []
    for i in range(1, floor(k/2-1)+1):
        G = F^(k+2)*(1-Lam_Series)*(Lam_Series)^i
        print (f"Power expansion up to q^{n} for G_{i}, k={k}\n")
        print (G.O(n))
        print ("\n")
        L.append(G)
    return L
def Gamma14_get_qexp(k, n=15):
    """
    Given k an even integer,
    Prints the q-expansion of all the basis of
    S_k+2(Gamma1(4)), for powers of q up to n
    """
    L = CuspForms(Gamma1(4),k+2).basis()
    i = 0
    for exp in L:
        print (f"Gamma1({4}) Basis #{i+1} of S_{k+2}: up to q^{n}")
        print(exp.qexp(n))
        print("\n")
        i += 1
    return L
```

Listing 4: Establishing Linear Relationship between functions $G_i^{(k)}$ and basis of $\mathbb{S}_{k+2}(\Gamma_1(4))$.

Running

```
Ex2=sqrt(x^3+x^2-t)
show(find_Picard(Ex2))
```

Outputs

$$\left[B = \frac{2\,(27\,t-2)}{27\,t^2-4\,t}, C = \frac{15}{4\,(27\,t^2-4\,t)}\right].$$

We can use any `SageMath` command, or the method of Frobenius, such as

```
y = function('y')(t)
de = diff(y, t, 2) + (2*(27*t - 2)/(27*t^2 - 4*t))*diff(y, t, 1)+  15/(4*
    (27*t^2 - 4*t))*y == 0
print(desolve(de, y, contrib_ode=True, show_method=True))
```

to find the solution and the basis as $\pi_1(t) = {}_2F_1\left(\frac{1}{6}, \frac{5}{6}; 1; \frac{27}{4}t\right), \pi_2(t) = i \cdot {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; 1 - \frac{27}{4}t\right)$. Then a similar argument to theorem 3.1 will go through, and produce the two monodromy matrices:

$$\rho(\gamma_0) = \begin{pmatrix} 1 & 0 \\ e^{\frac{\pi i}{3}} & 1 \end{pmatrix}, \qquad \rho(\gamma_1) = \begin{pmatrix} 1 & e^{\frac{2\pi i}{3}} \\ 0 & 1 \end{pmatrix}$$

hence the image of ρ is conjugate to $\mathrm{SL}_2(\mathbb{Z})$(proof, just find the matrix).

The respective modular form is a modular form called x_2 introduced in an article by Shen [2013]. We can get its q-expansion by the following `SageMath` code in Listing 5. This requires the definition of `Q_Ring` in Listing 4.

```
E_4_list = list(eisenstein_series_qexp(4, 50, normalization ='constant'))
E_4_list = [item for items in zip(E_4_list, [0] * len(E_4_list)) for item in items]
E_6_list = list(eisenstein_series_qexp(6, 50, normalization ='constant'))
E_6_list = [item for items in zip(E_6_list, [0] * len(E_6_list)) for item in items]
E_4 = Q_Ring(E_4_list, 99)

E_6 = Q_Ring(E_6_list, 99)

x_2 = (1/2)*(1- (E_6)/((E_4)**(3/2)))
```

Listing 5: The variable `x_2` which holds the first 99 terms of the q-expansion of x_2.

However, the author is struggling to find a similar expression analogous to $G_i^{(k)}$ in equation (6) in this case. One might expect it to be related to the Wronskian of the Picard-Fuchs Equation.

5.2 Conclusion and Further Work

In equation (8) we established explicit linear transformation between the basis and functions $G_i^{(k)}$ for the Legendre Family. Given theorem 4.4 *a priori*, the author wishes to automate the process and extract all the coefficients for the transformation, and understand if there are any relations between different k.

There is also an ODE with $p = 2$, but the author does not know whether it can be changed into the normal form without other technical concerns.

Bibliography

Lars Valerian Ahlfors. *Complex Analysis*. Mcgraw-Hill, 1979.

Harry Bateman. *Higher Transcendental Functions, Volume I*. McGraw-Hill, 1953. URL https://resolver.caltech.edu/CaltechAUTHORS:20140123-104529738.

Simon K Donaldson. *Riemann surfaces*. Oxford University Press, 2011.

C. Douglas Haessig. Kloosterman sums and hecke polynomials in characteristics 2 and 3. *Finite Fields and Their Applications*, 76:101896, 2021. ISSN 1071-5797. doi: https://doi.org/10.1016/j.ffa.2021.101896. URL https://www.sciencedirect.com/science/article/pii/S1071579721000903.

Allen Hatcher. Algebraic topology, 2002. URL https://pi.math.cornell.edu/~hatcher/AT/AT.pdf.

Katsunori Iwasaki, Hironobu Kimura, Shun Shimemura, and Masaaki Yoshida. *From Gauss to Painlevé*. Vieweg+Teubner Verlag, 06 2012.

Michio Kuga. *Galois' Dream: Group Theory and Differential Equations*. Birkhauser, 1993.

Ryan Lam. Elliptic surfaces and the picard-fuchs equation, 02 2023. URL https://ryanlamhk.files.wordpress.com/2023/02/picardfuchs-1.pdf.

Christian Schnell. On computing picard-fuchs equation, 2004. URL https://www.math.stonybrook.edu/~cschnell/pdf/notes/picardfuchs.pdf.

Li-Chien Shen. A note on Ramanujan's identities involving the hypergeometric function ${}_2f_1(\frac{1}{6},\frac{5}{6};1;z)$. *The Ramanujan Journal*, 30(2):211–222, February 2013. ISSN 1572-9303. doi: 10.1007/s11139-011-9360-8. URL https://doi.org/10.1007/s11139-011-9360-8.

Joseph H Silverman. *The Arithmetic of Elliptic Curves*. Springer, 01 2014.

Michael Somos. A115977 - oeis, 2015. URL https://oeis.org/A115977.

Andrew Wiles. Modular elliptic curves and fermat's last theorem. *Annals of Mathematics*, 141(3): 443–551, 1995. ISSN 0003486X. URL http://www.jstor.org/stable/2118559.

Wenzhe Yang. Seiberg-Witten theory and modular lambda function. 2019.

Masaaki Yoshida. *Hypergeometric Functions, My Love*. Springer Science & Business Media, 06 2013.